

Table 66. WLCSP12 – Mechanical data

Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A ⁽²⁾	-	-	0.49	-	-	0.0193
A1	-	0.17	-	-	0.0067	-
A2	-	0.29	-	-	0.0114	-
A3 ⁽³⁾	-	0.025	-	-	0.0098	-
Ø b ⁽⁴⁾	0.21	0.24	0.27	0.0083	0.0094	0.0106
D	1.68	1.70	1.72	0.0661	0.0669	0.0677
E	1.41	1.42	1.43	0.0555	0.0559	0.0563
e	-	0.35	-	-	0.0138	-
e1	-	0.909	-	-	0.0358	-
e2	-	0.875	-	-	0.0344	-
F ⁽⁵⁾	-	0.409	-	-	0.0161	-
G ⁽⁵⁾	-	0.282	-	-	0.0111	-
N	12					
aaa	-	-	0.10	-	-	0.0039
bbb	-	-	0.10	-	-	0.0039
ccc ⁽⁶⁾	-	-	0.10	-	-	0.0039
ddd ⁽⁷⁾	-	-	0.05	-	-	0.0020
eee	-	-	0.05	-	-	0.0020

1. Values in inches are converted from mm and rounded to 4 decimal digits.
2. The maximum total package height is calculated by the RSS method (Root Sum Square) using nominal and tolerances values of A1 and A2.
3. Back side coating. Nominal dimension is rounded to the 3rd decimal place resulting from process capability.
4. Dimension is measured at the maximum bump diameter parallel to primary datum Z.
5. Calculated dimensions are rounded to the 3rd decimal place
6. Bump position designation per JESD 95-1, SPP-010. The tolerance of position that controls the location of the pattern of balls with respect to datums X and Y. For each ball there is a cylindrical tolerance zone ccc perpendicular to datum Z and located on true position with respect to datums X and Y as defined by e. The axis perpendicular to datum Z of each ball must lie within this tolerance zone.
7. The tolerance of position that controls the location of the balls within the matrix with respect to each other. For each ball there is a cylindrical tolerance zone ddd perpendicular to datum Z and located on true position as defined by e. The axis perpendicular to datum Z of each ball must lie within this tolerance zone. Each tolerance zone ddd in the array is contained entirely in the respective zone ccc above. The axis of each ball must lie simultaneously in both tolerance zones